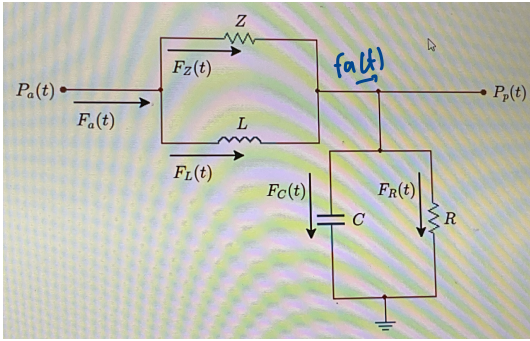




Ecuación principal

$$F_a(t) = F_z(t) + F_c(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z} \quad F_c(t) = \frac{dP_p(t)}{dt}$$

$$F_L(t) = \frac{1}{L} \int [P_a(t) - P_p(t)] dt \quad F_R(t) = \frac{P_p(t)}{R}$$

Procedimiento algebraico

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt = \frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{Ls} \right) P_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls} \right) P_p(s)$$

$$\frac{Ls + Z}{LZs} P_a(s) = \frac{CLs^2 + LZs + RLS + RZ}{RLZs} P_p(s)$$

Función de transferencia

$$\frac{P_P(s)}{P_A(s)} = \frac{\frac{Ls + R_z}{Lz s}}{\frac{CLs^2 + (Lz s + RL)s + Rz}{RLz s}}$$

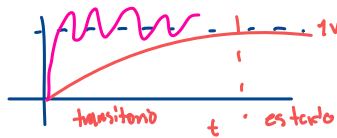
$$\frac{P_P(s)}{P_A(s)} = \frac{RL \cancel{z} s^2 + RL z^2 s}{CL^2 z s^4 + (L^2 z^2 s^2 + RL^2) s^2 \cancel{RL z^2 s}} \quad CL^2 z s$$

$$H_s = \frac{RLs + Rz}{CL Rz s^2 + (Lz + RL)s + Rz}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} \frac{1}{s} \left[1 - \frac{RLs^0 + Rz}{CL \cancel{Rz} s^2 + (Lz + RL)s + Rz} \right]$$

$$= 1 - \frac{Rz}{Rz} = 0$$



Estabilidad en lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_{1,2} = \frac{-(Lz + RL) \pm \sqrt{(Lz + RL)^2 - 4CLRz}}{2CLRz}$$

$$a = CLRz$$

$$b = Lz + RL$$

$$c = Rz$$

El sistema tiene una respuesta estable

porque $\text{Re}(\lambda_{1,2}) < 0$

10-Oct-2025

Modelo de ecuaciones integro-diferenciales

$$P_P(t) \left(\frac{1}{R} + \frac{1}{Z} \right) = \frac{P_A(t)}{Z} + \frac{1}{Z} \int [P_A(t) - P_P(t)] dt - C \frac{dP_P(t)}{dt}$$

$$P_P(t) = \left(\frac{P_A(t)}{Z} + \frac{1}{Z} \int [P_A(t) - P_P(t)] dt - C \frac{dP_P(t)}{dt} \right) \frac{ZR}{Z+R}$$

